

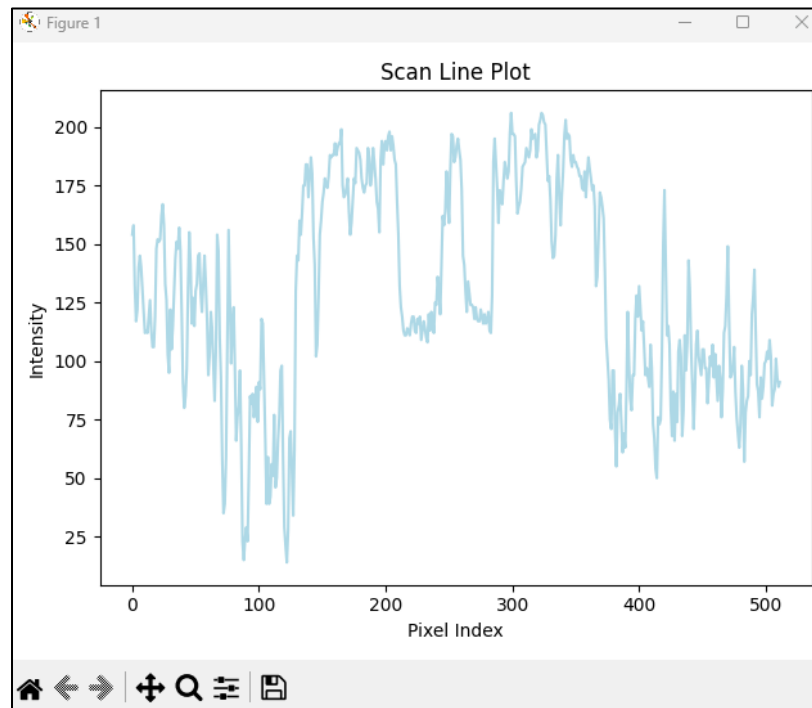
Assignment 1

This is an individual assignment. This assignment is marked out of 5 points.

Due Date: October 2 Friday, 2026, 11:59PM

1. Extracting intensity scan lines from grayscale images.

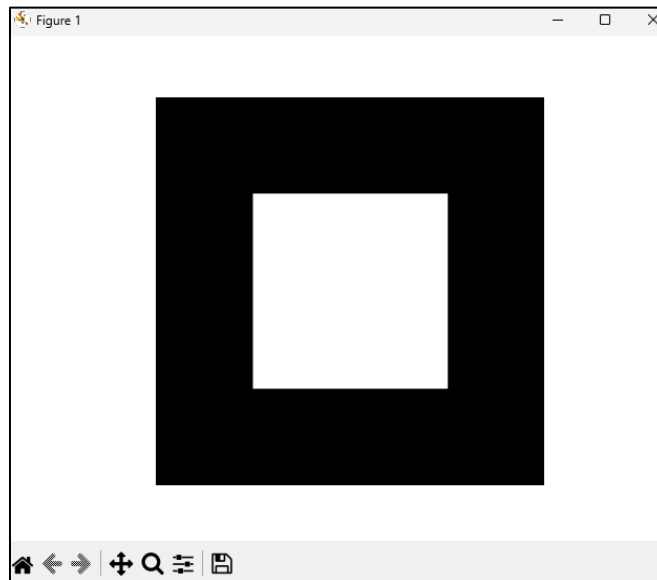
- (a) [1 point] Write a function, `scanLine4e(f, l, loc)`, where f is a grayscale image, l is an integer, and loc is a character string with value 'row' or 'col' indicating that l is either a row or column number in image f . The output, s , is a vector containing the pixel values along the specified row or column in f .
- (b) [1 point] Read the image "Baboon.png", convert it to a grayscale image and get a horizontal intensity scan line in the middle of the image, and plot the scan line.



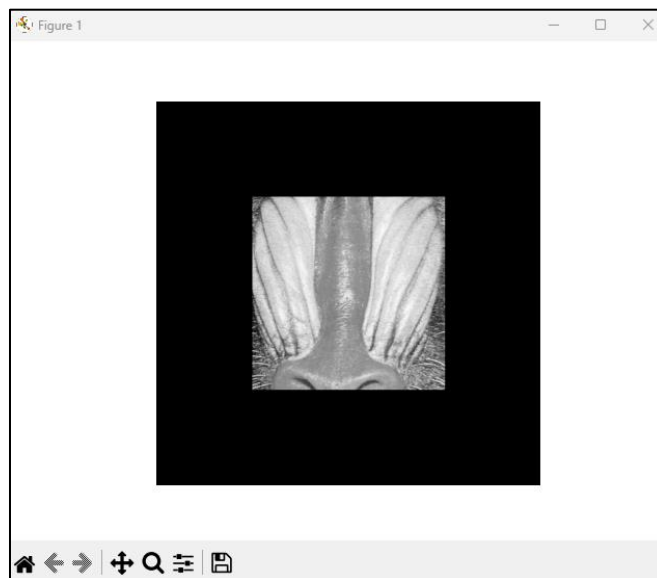
2. Rectangular binary masks:

- (a) [1 point] Write a function `mask4e(M, N, rUL, cUL, rLR, cLR)` for creating a binary mask of size $M \times N$ with 1's in the rectangular region defined by upper-left row and column coordinates rUL, cUL , and lower-right coordinates rLR, cLR , respectively. All coordinates are inclusive. The rest of the $M \times N$ mask should be 0's. Your function should contain a check to make sure the specified region of 1's does not exceed the dimensions of the $M \times N$ region.

- (b) [1 point] Read the image “BaboonGray.jpg” and generate a square mask whose sides are one-half the size of the image in both directions. The mask should be centered on the image.



- (c) [0.5 point] Confirm that the number of the 1-valued elements in the mask is correct.
(d) [0.5 point] Apply your mask to image “BaboonGray.jpg” and display your result.



Requirements:

- Only the following packages are permitted: OpenCV-Python, Matplotlib, NumPy, Math
- Use Python version 3.10+

Submission:

Submit everything as one zip file to Canvas.